

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415404
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Kim; Hak Kyu et al.

Vehicular air conditioning system

Abstract

The present invention relates to a vehicular air conditioning system and provides a vehicular air conditioning system capable of preventing overheating of an inverter that controls the rotation speed of an electric compressor without cutting off the inverter, and consequently preventing the stoppage of an air conditioner due to the cut-off of the inverter and the resultant stoppage of cooling of a passenger room. The vehicular air conditioning system includes an electric compressor, an inverter configured to control a rotation speed of the electric compressor, and a control part configured to execute control so that a refrigerant flow rate on an inlet side of the electric compressor increases when a temperature of the inverter is equal to or higher than a preset overheating warning temperature.

Inventors:	Kim; Hak Kyu (Daejeon, KR), Kim; In Hyeok (Daejeon, KR), Yoo; Sang Jun (Daejeon, KR), Jung; Hong Rae (Daejeon, KR)
Applicant:	Hanon Systems (Daejeon, KR)
Family ID:	1000008820961
Assignee:	HANON SYSTEMS (Daejeon, KR)
Appl. No.:	18/569954
Filed (or PCT Filed):	September 23, 2022
PCT No.:	PCT/KR2022/014268
PCT Pub. No.:	WO2023/058971
PCT Pub. Date:	April 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240278622 A1	Aug. 22, 2024

Foreign Application Priority Data

KR	10-2021-0132460	Oct. 06, 2021
KR	10-2022-0107316	Aug. 26, 2022

Publication Classification

Int. Cl.: B60H1/32 (20060101)

U.S. Cl.:

CPC B60H1/3216 (20130101); B60H1/3222 (20130101);

Field of Classification Search

CPC: F25B (2600/021); F25B (2600/025); F25B (2700/21154); F25B (2700/2117); F25B (49/025); F25B (49/005); B60H (1/3216); B60H (1/3222)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8459053	12/2012	Pham	62/208	F04C 28/28
2002/0108384	12/2001	Higashiyama	62/183	B60H 1/3222
2005/0210897	12/2004	Oomura	62/239	F25B 49/005
2016/0134226	12/2015	Najima	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20100027573	12/2009	KR	N/A
20110098315	12/2010	KR	N/A
20120061467	12/2011	KR	N/A
20120095735	12/2011	KR	N/A
20120110858	12/2011	KR	N/A

OTHER PUBLICATIONS

International Search Report issued in PCT/KR2022/014268 on Jan. 13, 2023. cited by applicant

Primary Examiner: Norman; Marc E

Attorney, Agent or Firm: NORTON ROSE FULBRIGHT US LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This patent application is a national phase under 35 U.S.C. § 371 of International Patent Application No. PCT/KR2022/014268 filed Sep. 23, 2022, which claims the benefit of priority

from Korean Patent Application Nos. 10-2021-0132460 filed Oct. 6 2021, and 10-2022-0107316 filed Aug. 26, 2022, each of which is hereby incorporated herein by reference in its entirety for all purposes.

TECHNICAL FIELD

(2) The present invention relates to a vehicular air conditioning system, and more particularly, a vehicular air conditioning system capable of preventing overheating of an inverter that controls the rotation speed of an electric compressor without cutting off the inverter, and consequently preventing the stoppage of an air conditioner due to the cut-off of the inverter and the resultant stoppage of cooling of a passenger room.

BACKGROUND ART

(3) In a hybrid vehicle and an electric vehicle (hereinafter collectively referred to as “vehicle”), all devices in the vehicle use electricity as a power source. In particular, an air conditioning system for adjusting the temperature inside the vehicle also uses electricity as a power source.

(4) As shown in FIG. 1, such an electrically-powered air conditioning system is equipped with an electric compressor **10** for compressing a refrigerant to a high temperature and high pressure state.

(5) The electric compressor **10** includes a compression part **12** for compressing a refrigerant, and a motor part **14** for driving the compression part **12**.

(6) In this electric compressor **10**, when the motor part **14** is operated by the electricity applied thereto, the compression part **12** is driven. When the compression part **12** is driven, the refrigerant on the evaporator **20** side is introduced into an inlet **16** and is compressed to a high temperature and high pressure by the compression part **12**.

(7) The compressed high-temperature high-pressure refrigerant is discharged toward the condenser **22** through the discharge port **18**.

(8) Meanwhile, the rotation speed of the electric compressor **10** is adjusted by an inverter **30**, and the electric compressor **10** having the adjusted rotation speed adjusts the flow rate of the refrigerant flowing toward a condenser **22**, an expansion valve **24**, and an evaporator **20**.

(9) The inverter **30** includes an inverter housing **32**, a circuit board **34** built into the inverter housing **32**, and a plurality of circuit elements **36** installed on the circuit board **34**.

(10) The inverter **30** applies or cuts off electric power supplied to the motor part **14** and controls the rotation speed of the motor part **14**.

(11) In particular, when an evaporator target temperature is set according to the cooling and heating load in the passenger room, i.e., the temperature conditions inside and outside the passenger room and the user-set temperature, the rotation speed of the motor part **14** is controlled in accordance with the set evaporator target temperature.

(12) Therefore, the electric compressor **10** can discharge an optimal amount of refrigerant suitable for the evaporator target temperature. As a result, the temperature of the evaporator **20** can be controlled to match the evaporator target temperature.

(13) Meanwhile, the inverter **30** generates heat during the control process of the electric compressor **10**. In particular, among the circuit elements **36** built into the inverter **30**, an IGBT (Insulated Gate Bipolar Transistor) element generates heat at a high temperature. This heat generation may cause the inverter **30** to be overheated and subjected to damage or failure.

(14) As a solution to the overheating of the inverter **30**, there is known a cooling technique that makes use of a refrigerant at the inlet **16** of the electric compressor **10**.

(15) In this technique, the inverter **30** is integrally installed on the side of the refrigerant inlet **16** of the electric compressor **10**, so that the relatively low temperature refrigerant sucked into the refrigerant inlet **16** of the electric compressor **10** indirectly contacts the inverter **30**.

(16) Therefore, heat exchange can occur between the low-temperature refrigerant at the inlet **16** and the inverter **30**, thereby cooling inverter **30**.

(17) However, the conventional air conditioning device has a disadvantage that the cooling efficiency of the inverter **30** is reduced when the flow rate of the refrigerant sucked into the inlet **16**

of the electric compressor **10** is low. Therefore, the inverter **30** is highly likely to be overheated.
(18) In consideration of this, there is available a technique for preventing the overheating of the inverter **30**.

(19) This technology fundamentally prevents the overheating of the inverter **30** by cutting off the inverter **30** when the temperature of the inverter **30** is equal to or higher than a preset overheating reference temperature. Therefore, damage and failure of the inverter **30** due to the overheating are prevented.

(20) The technique for cutting off the inverter **30** is configured to cut off the inverter **30** whenever the inverter **30** is overheated. Therefore, the air conditioning system is stopped whenever the inverter **30** is overheated. Therefore, the cooling and heating in the passenger room are stopped.

(21) In particular, the overheating of the inverter **30** may frequently occur in the summer when the temperature is high. This may frequently stop the air conditioning system, thereby significantly reducing the comfort in the passenger room.

SUMMARY

(22) In view of the problems inherent in the related art, it is an object of the present invention to provide a vehicular air conditioning system capable of preventing the overheating of an inverter without cutting off the inverter.

(23) Another object of the present invention is to provide a vehicular air conditioning system capable of preventing the stoppage of an air conditioner due to the cut-off of an inverter and the resultant stoppage of cooling of a passenger room.

(24) A further object of the present invention is to provide a vehicular air conditioning system capable of significantly improving the comfort in a passenger room.

(25) In order to achieve these objects, there is provided a vehicular air conditioning system, including: an electric compressor; an inverter configured to control a rotation speed of the electric compressor; and a control part configured to execute control so that a refrigerant flow rate on an inlet side of the electric compressor increases when a temperature of the inverter is equal to or higher than a preset overheating warning temperature.

(26) The control part may be configured to execute control so that when the temperature of the inverter is equal to or higher than the overheating warning temperature, an evaporator target temperature is variably adjusted according to the temperature of the inverter to variably adjust the rotation speed of the electric compressor.

(27) The inverter may be installed on the inlet side of the electric compressor and cooled by an inlet side refrigerant, and the control part may be configured to control the rotation speed of the electric compressor according to the temperature of the inverter so that the refrigerant flow rate on the inlet side of the electric compressor for cooling the inverter is variably adjusted depending on the temperature of the inverter.

(28) As the temperature of the inverter increases toward an overheating reference temperature set higher than the overheating warning temperature, the control part may gradually lower the evaporator target temperature in inverse proportion thereto, and as the temperature of the inverter approaches the overheating reference temperature from the overheating warning temperature, the control part may execute control to gradually increase the rotation speed of the electric compressor so that the refrigerant flow rate on the inlet side of the electric compressor gradually increases as the temperature of the inverter approaches the overheating reference temperature.

(29) The control part may store table values of evaporator target temperature drop amounts for respective inverter temperatures, and when the temperature of the inverter increases toward the overheating reference temperature, the control part may variably adjust the evaporator target temperature according to the temperature of the inverter based on the table values of evaporator target temperature drop amounts for respective inverter temperatures.

(30) When the temperature of the inverter increases toward the overheating reference temperature, the control part may detect an evaporator target temperature drop amount corresponding to the

temperature of the inverter from the table values of evaporator target temperature drop amounts for respective inverter temperatures, and the control part may be configured to calculate a final evaporator target temperature (T3) by processing a detected evaporator target temperature drop amount (T2) and an evaporator target temperature (T1) preset according to temperature conditions inside and outside a passenger room and a user-set temperature by equation (1) below, and change the calculated final evaporator target temperature (T3) to the evaporator target temperature: final evaporator target temperature (T3)=preset evaporator target temperature (T1)–evaporator target temperature drop amount (T2) corresponding to inverter temperature . . . (1).

(31) According to the vehicular air conditioning system of the present invention, as the temperature of the inverter approaches the overheating reference temperature, the evaporator target temperature is gradually lowered to gradually increase the refrigerant discharge amount of the electric compressor. Therefore, as the temperature of the inverter approaches the overheating reference temperature, it is possible to increase the refrigerant flow rate on the inlet side of the electric compressor and to gradually increase the cooling efficiency of the inverter through the increase in the refrigerant flow rate on the inlet side.

(32) Further, since the cooling efficiency of the inverter can be gradually increased by increasing the refrigerant flow rate on the inlet side of the electric compressor, unlike the prior art, the overheating of the inverter can be prevented without cutting off the inverter.

(33) In addition, since the overheating of the inverter can be prevented without cutting off the inverter, it is possible to prevent the stoppage of the air conditioner due to the cut-off of the inverter and the resultant stoppage of cooling of the passenger room, thereby significantly improving the comfort in the passenger room.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram schematically showing a conventional vehicular air conditioning system.

(2) FIG. 2 is a diagram showing the configuration of a vehicular air conditioning system according to the present invention.

(3) FIG. 3 is a flowchart showing an operation example of the vehicular air conditioning system according to the present invention.

DETAILED DESCRIPTION

(4) A preferred embodiment of a vehicular air conditioning system according to the present invention will now be described in detail with reference to the accompanying drawings.

(5) Prior to describing features of the vehicular air conditioning system according to the present invention, an electric compressor of a vehicular air conditioning system will be described briefly with reference to FIG. 2.

(6) The electric compressor **10** includes a compression part **12** and a motor part **14**. When the compression part **12** is driven by the motor part **14**, the refrigerant on the evaporator **20** side is introduced into an inlet **16** and is compressed to a high temperature and high. The compressed high-temperature high-pressure refrigerant is discharged toward a condenser **22** through a discharge port **18**.

(7) The rotation speed of the electric compressor **10** is adjusted by an inverter **30**, and the electric compressor **10** having the adjusted rotation speed adjusts the flow rate of the refrigerant flowing toward a condenser **22**, an expansion valve **24**, and an evaporator **20**.

(8) The inverter **30** includes an inverter housing **32**, a circuit board **34** built into the inverter housing **32**, and a plurality of circuit elements **36** installed on the circuit board **34**.

(9) The inverter **30** applies or cuts off electric power supplied to the motor part **14** and controls the rotation speed of the motor part **14**.

(10) In particular, when an evaporator target temperature is set in a control part **40** according to the temperature conditions inside and outside the passenger room and the user-set temperature, the rotation speed of the motor part **14** is controlled in accordance with the set evaporator target temperature of the control part **40**.

(11) Therefore, the electric compressor **10** can discharge an optimal amount of refrigerant suitable for the evaporator target temperature. As a result, the temperature of the evaporator **20** can be controlled to match the evaporator target temperature.

(12) In this regard, the control part **40** is equipped with a microprocessor, and may be installed integrally on the circuit board **34** within the inverter **30**, or may be configured separately from the inverter **30** as the case may be.

(13) Next, the features of the vehicular air conditioning system according to the present invention will be described in detail with reference to FIGS. **2** and **3**.

(14) Referring first to FIG. **2**, the air conditioning system of the present invention is provided with the control part **40**. The control part **40** is configured to variably adjust the evaporator target temperature according to the sensing value of an inverter temperature sensor **42** that detects the temperature of the inverter **30**.

(15) Specifically, the control part **40** is configured to variably adjust the evaporator target temperature according to the inputted inverter temperature when temperature data of the inverter **30** is inputted from the inverter temperature sensor **42**.

(16) In particular, as the inverter temperature increases to a temperature approaching a preset overheating reference temperature, the evaporator target temperature is gradually lowered by a preset temperature in inverse proportion to the increase in the inverter temperature.

(17) For example, if the overheating reference temperature of the inverter **30** is 120 degrees C., each time when the temperature of the inverter **30** increases toward the overheating reference temperature (120 degrees C.) by a preset temperature (2 degrees C.) from a specific overheating warning temperature (114 degrees C.) below the overheating reference temperature of 120 degrees C., the evaporator target temperature is gradually lowered by a preset temperature (1 degrees C.) in inverse proportion to the increase in the temperature of the inverter.

(18) Therefore, when the temperature of the inverter **30** gradually increases from an overheating warning temperature to a temperature approaching the overheating reference temperature set higher than the overheating warning temperature, the inverter **30** gradually increases the rotation speed of the electric compressor **10** correspondingly.

(19) Thus, as the temperature of the inverter **30** gradually increases from the overheating warning temperature to a temperature approaching the overheating reference temperature, the refrigerant discharge flow rate of the electric compressor **10** is gradually increased in proportion thereto.

(20) As a result, as the temperature of the inverter **30** increases to a temperature approaching the overheating reference temperature, the amount of the refrigerant returned to the inlet **16** of the electric compressor **10** is also gradually increased in proportion thereto.

(21) Accordingly, as the temperature of the inverter **30** increases to a temperature approaching the overheating reference temperature, the heat exchange rate between the refrigerant sucked into the inlet **16** of the electric compressor **10** and the inverter **30** increases.

(22) Therefore, as the temperature of the inverter **30** increases to a temperature approaching the overheating reference temperature, the cooling efficiency of the inverter **30** is increased to prevent the overheating of the inverter **30**.

(23) Consequently, as the temperature of the inverter **30** approaches the overheating reference temperature, the control part **40** gradually enhances the cooling efficiency of the inverter **30** through the refrigerant on the inlet side of the electric compressor **10**, thereby preventing the overheating of the inverter **30**.

(24) In particular, the overheating of the inverter **30** can be prevented without cutting off the inverter **30**. Therefore, it is possible to prevent the stoppage of the air conditioner due to the cut-off

of the inverter **30** and the resultant stoppage of cooling of the passenger room, thereby significantly improving the comfort in the passenger room.

(25) Meanwhile, in some cases, the control part **40** may store target evaporator temperature drop amounts for respective inverter temperatures, as shown in Table 1 below.

(26) The target evaporator temperature drop amounts for respective inverter temperatures are stored in the control part **40** over a range from the overheating warning temperature of the inverter **30**, for example, 114 degrees C., to the overheating reference temperature or higher, for example, 126 degrees C., which is higher than 120 degrees C.

(27) By using the stored evaporator target temperature drop amounts for respective inverter temperatures, the evaporator target temperature is varied according to the temperature of the inverter **30**. Thus, the control valve of the inverter **30** for the electric compressor **10** is variably controlled according to the temperature of the inverter **30** so that the refrigerant discharge amount of the electric compressor **10** can be changed.

(28) TABLE-US-00001 TABLE 1 Table values of evaporator target temperature drop amounts for respective inverter temperatures

Evaporator target temperature drop (degrees C.)	Inverter temperature amount (degrees C.)
A (Overheating warning temperature)	112.0
B	114.0
C	116.0
D (Overheating reference temperature)	118.0
	120.0
	122.0
	124.0
	126.0

(29) In this regard, the control part **40** may change the evaporator target temperature based on the table values of evaporator target temperature drop amounts for respective inverter temperatures only under a condition in which the temperature of the inverter **30** is equal to or higher than the preset specific overheating warning temperature set to be lower than the overheating reference temperature, for example, 120 degrees C. or lower, for example, a condition in which the temperature of the inverter **30** increases to 114 degrees C. or higher.

(30) When changing the evaporator target temperature using the stored evaporator target temperature drop amounts for respective inverter temperatures, the control part **40** is configured to calculate a final evaporator target temperature (T3) by subtracting an evaporator target temperature drop amount (T2) corresponding to the inverter temperature from the evaporator target temperature (T1) preset according to the temperature conditions inside and outside the passenger room and the user-set temperature as in equation (1) below, and change the calculated final evaporator target temperature (T3) to the evaporator target temperature:

final evaporator target temperature (T3)=preset evaporator target temperature (T1)–evaporator target temperature drop amount (T2) corresponding to inverter temperature (1).

(31) As a result, when the temperature of the inverter **30** increases to the overheating warning temperature or higher, the control part **40** variably set the evaporator target temperature by compensating for the evaporator target temperature drop amount corresponding to the temperature of the inverter **30** temperature to the preset evaporator target temperature.

(32) Meanwhile, it is preferable that the evaporator target temperature drop amounts for respective inverter temperatures stored in the control part **40** increase linearly in proportion to the temperature of the inverter **30**.

(33) In particular, it is preferable that, among the evaporator target temperature drop amounts for respective inverter temperatures, the evaporator target temperature drop amount corresponding to the overheating warning temperature or higher and the overheating reference temperature or lower increases linearly in proportion to the temperature of the inverter **30**.

(34) Accordingly, as the temperature of the inverter **30** increases, the evaporator target temperature drop amount can also be gradually increased in proportion thereto. As a result, as the inverter **30** is overheated, the evaporator target temperature gradually decreases, making it possible to actively respond to the overheating of the inverter **30**.

(35) Meanwhile, among the evaporator target temperature drop amounts for respective inverter temperatures stored in the control part **40**, the evaporator target temperature drop amount corresponding to the overheating reference temperature of the inverter **30** or higher is preferably

maintained at a constant level.

(36) Accordingly, the control part **40** is configured to change the evaporator target temperature based on the evaporator target temperature drop amount of the constant level when the temperature of the inverter **30** is equal to or higher than the overheating reference temperature.

(37) The reason for adopting this configuration is that if the evaporator target temperature drop amount is made larger and the evaporator target temperature is lowered too much, an icing phenomenon may occur on the evaporator **20** side.

(38) Referring again to FIG. 2, when the temperature of the inverter **30** is equal to or higher than the overheating warning temperature, the control part **40** may execute control to increase the minimum rotation speed (RPM) of the electric compressor **10** so that the refrigerant flow rate at the inlet **16** of the electric compressor **10** can increase.

(39) The rotation speed of the electric compressor **10** is controlled according to the temperature of the evaporator **20**. At this time, there is a minimum rotation speed reference that prevents the rotation speed of the electric compressor **10** from falling below a predetermined rotation speed.

(40) Therefore, if the minimum rotation speed of the electric compressor **10** is increased based on this reference, the rotation speed of the electric compressor **10** increases even at the same evaporator **20** temperature. Thus, the flow rate of the refrigerant sucked at the inlet **16** of the electric compressor **10** increases to thereby improve the cooling efficiency of the inverter **30**.

(41) In some cases, when the temperature of the inverter **30** is equal to or higher than the overheating warning temperature, the control part **40** may execute control to increase the rotation speed (RPM) of the electric compressor **10** to a preset value, so that the flow rate of the refrigerant at the inlet **16** of the electric compressor **10** can increase.

(42) Next, an operational example of the vehicular air conditioning system of the present invention having such a configuration will be described with reference to FIGS. 2 and 3.

(43) Referring first to FIG. 3, when the air conditioning system is turned on (S101), the temperature of the inverter **30** is detected (S103).

(44) When the detection of temperature of the inverter **30** is completed, the control part **40** determines whether the detected temperature of the inverter **30** is equal to or higher than a preset overheating warning temperature, for example, 114 degrees C. (S105).

(45) As a result of the determination, if the detected temperature of the inverter **30** is equal to or higher than the preset overheating warning temperature, the control part **40** determines that there is a high risk of overheating of the inverter **30**.

(46) According to this determination, the control part **40** detects the evaporator target temperature drop amount corresponding to the current temperature of the inverter **30** from a pre-stored evaporator target temperature drop amount table as indicated in Table 1 above (S107).

(47) When the detection of the evaporator target temperature drop amount is completed, the detected evaporator target temperature drop amount (T2) and the preset evaporator target temperature (T1) are processed using equation (1) above to calculate a final evaporator target temperature (T3) (S109).

(48) When the calculation of the final evaporator target temperature (T3) is completed, the inverter **30** is able to control the electric compressor **10** according to the calculated final evaporator target temperature (T3) (S111).

(49) Then, when the temperature of the inverter **30** gradually increases beyond the overheating warning temperature to the overheating reference temperature, the rotation speed of the electric compressor **10** gradually increases in response thereto.

(50) Therefore, as the temperature of the inverter **30** gradually increases to the overheating reference temperature, the refrigerant discharge flow rate of the electric compressor **10** gradually increases in proportion thereto, so that the cooling performance of the inverter **30** through the refrigerant on the inlet side of the electric compressor **10** can gradually increase. Thus, the overheating of the inverter **30** is prevented.

(51) According to the vehicular air conditioning system of the present invention having such a configuration, as the temperature of the inverter **30** approaches the overheating reference temperature, the evaporator target temperature is gradually lowered to gradually increase the refrigerant discharge amount of the electric compressor **10**. Therefore, as the temperature of the inverter **30** approaches the overheating reference temperature, it is possible to increase the refrigerant flow rate on the inlet side of the electric compressor **10** and to gradually increase the cooling efficiency of the inverter **30** through the increase in the refrigerant flow rate on the inlet side.

(52) Further, since the cooling efficiency of the inverter **30** can be gradually increased by increasing the refrigerant flow rate on the inlet side of the electric compressor **10**, unlike the prior art, the overheating of the inverter **30** can be prevented without cutting off the inverter **30**.

(53) In addition, since the overheating of the inverter **30** can be prevented without cutting off the inverter **30**, it is possible to prevent the stoppage of the air conditioner due to the cut-off of the inverter **30** and the resultant stoppage of cooling of the passenger room, thereby significantly improving the comfort in the passenger room.

(54) While the preferred embodiments of the present invention have been described above, the present invention is not limited to the above-described embodiments. Various modifications and changes may be made without departing from the scope and spirit of the present invention defined in the claims.

Claims

1. A vehicular air conditioning system, comprising: an electric compressor; an inverter configured to control a rotation speed of the electric compressor; and a control part configured to execute control so that a refrigerant flow rate on an inlet side of the electric compressor increases when a temperature of the inverter is equal to or higher than a preset overheating warning temperature, wherein the control part is configured to execute control so that when the temperature of the inverter is equal to or higher than the overheating warning temperature, an evaporator target temperature is variably adjusted according to the temperature of the inverter to variably adjust the rotation speed of the electric compressor.
2. The system of claim 1, wherein the inverter is installed on the inlet side of the electric compressor and cooled by an inlet side refrigerant, and the control part is configured to control the rotation speed of the electric compressor according to the temperature of the inverter so that the refrigerant flow rate on the inlet side of the electric compressor for cooling the inverter is variably adjusted depending on the temperature of the inverter.
3. The system of claim 2, wherein as the temperature of the inverter increases toward an overheating reference temperature set higher than the overheating warning temperature, the control part gradually lowers the evaporator target temperature in inverse proportion thereto, and as the temperature of the inverter approaches the overheating reference temperature from the overheating warning temperature, the control part executes control to gradually increase the rotation speed of the electric compressor so that the refrigerant flow rate on the inlet side of the electric compressor gradually increases as the temperature of the inverter approaches the overheating reference temperature.
4. The system of claim 3, wherein the control part stores table values of evaporator target temperature drop amounts for respective inverter temperatures, and when the temperature of the inverter increases toward the overheating reference temperature, the control part variably adjusts the evaporator target temperature according to the temperature of the inverter based on the table values of evaporator target temperature drop amounts for respective inverter temperatures.
5. The system of claim 4, wherein when the temperature of the inverter increases toward the overheating reference temperature, the control part detects an evaporator target temperature drop

amount corresponding to the temperature of the inverter from the table values of evaporator target temperature drop amounts for respective inverter temperatures, and the control part is configured to calculate a final evaporator target temperature ($T3$) by processing a detected evaporator target temperature drop amount ($T2$) and an evaporator target temperature ($T1$) preset according to temperature conditions inside and outside a passenger room and a user-set temperature by equation (1) below, and change the calculated final evaporator target temperature ($T3$) to the evaporator target temperature:

final evaporator target temperature ($T3$)=preset evaporator target temperature ($T1$)−evaporator target temperature drop amount ($T2$) corresponding to inverter temperature (1).

6. The system of claim 5, wherein the evaporator target temperature drop amounts for respective inverter temperatures stored in the control part increases linearly in proportion to the temperature of the inverter.

7. The system of claim 5, wherein among the evaporator target temperature drop amounts for respective inverter temperatures stored in the control part, the evaporator target temperature drop amount corresponding to the overheating warning temperature or higher and the overheating reference temperature or lower increases linearly in proportion to the temperature of the inverter.

8. The system of claim 7, wherein among the evaporator target temperature drop amounts for respective inverter temperatures stored in the control part, the evaporator target temperature drop amount corresponding to the overheating reference temperature of the inverter or higher is maintained at a constant level, and the control part is configured to change the evaporator target temperature based on the evaporator target temperature drop amount of the constant level when the temperature of the inverter is equal to or higher than the overheating reference temperature.

9. The system of claim 4, wherein the control part is configured to change the evaporator target temperature based on the table values of evaporator target temperature drop amounts for respective inverter temperatures only under a condition in which the temperature of the inverter is equal to or higher than the overheating warning temperature set to be lower than the overheating reference temperature.

10. The system of claim 1, wherein when the temperature of the inverter is equal to or higher than the overheating warning temperature, the control part is configured to increase the rotation speed (RPM) of the electric compressor so that the refrigerant flow rate on the inlet side of the electric compressor increases.
